Title	**HRAM Analysis of 24 Oxadiazepine Compounds**
Sample Identification:	See Appendix 3
Project Code	BD70378
MS# / ASA#:	MS 26-018 / RMS00766
Submitted by:	Jon Reeves
Authored by:	Lifen Wu
Reviewed by:	Jordan Hartig and Scott Pennino
Test Laboratory:	Mass Spectrometry, Material and Analytical Sciences Department
EWBc Reference:	1594771
Date of Report:	08Apr2026

1. Introduction

The objective of this study was to provide high resolution accurate mass for 24 oxadiazepine compounds.

2. Experimental

2.1. MS Instrument and Method

See Appendix 1 for details.

2.2. Samples and Sample Preparation

See Appendix 2 and 3 for detailed submission form and structure list, respectively.

Sample Preparation:

Samples were analyzed "neat" by placing them directly in the DART Source ion stream.

3. Results and Discussion

Table 1 summarizes the high resolution accurate mass obtained for the 24 submitted samples. The mass spectrum of each compound along with an isotopic model is displayed in Figures 1-24.

Table 1: High Resolution Accurate Mass Results

Sample Name	Measured Mass	Theoretical Mass	Formula $[M+H]^+$	Mass Error (PPM)
MR105	339.0549	339.0551	$[C_{16}H_8N_2OF_5]^+$	-0.768
MR56	416.9655	416.9656	$[C_{16}H_7N_2OBrF_5]^+$	-0.439
MR62	355.0498	355.0500	$[C_{16}H_8N_2O_2F_5]^+$	-0.803
MR63	338.0709	338.0711	$[C_{16}H_9N_3F_5]^+$	-0.724
MR89	338.0597	338.0599	$[C_{17}H_9NOF_5]^+$	-0.537
MR65	303.0550	303.0551	$[C_{13}H_8N_2OF_5]^+$	-0.298
MR64	321.0290	321.0293	$[C_{12}H_6N_2O_3F_5]^+$	-0.871
MR73	357.0291	357.0293	$[C_{15}H_6N_2O_3F_5]^+$	-0.475
MR99	267.0926	267.0928	$[C_{16}H_{12}N_2OF]^+$	-0.666
MR92	285.0832	285.0834	$[C_{16}H_{11}N_2OF_2]^+$	-0.547
MR88	312.0776	312.0779	$[C_{16}H_{11}N_3O_3F]^+$	-0.916
MR116	362.9937	362.9939	$[C_{16}H_{10}N_2OBrF_2]^+$	-0.492
MR111	330.0683	330.0685	$[C_{16}H_{10}N_3O_3F_2]^+$	-0.497
MR113	321.0643	321.0646	$[C_{16}H_9N_2OF_4]^+$	-0.692
MR94	284.0879	284.0881	$[C_{17}H_{12}NOF_2]^+$	-0.975
MR108	319.0486	319.0489	$[C_{16}H_7N_2OF_4]^+$	-0.916
MR101	265.0769	265.0772	$[C_{16}H_{10}N_2OF]^+$	-0.897
MR59	396.9591	396.9594	$[C_{16}H_6N_2OBrF_4]^+$	-0.843
MR96	318.0533	318.0537	$[C_{17}H_8NOF_4]^+$	-0.985
MR115	301.0581	301.0583	$[C_{16}H_8N_2OF_3]^+$	-0.877
MR120	342.9874	342.9877	$[C_{16}H_9N_2OBrF]^+$	-0.788
MR119	310.0620	310.0622	$[C_{16}H_9N_3O_3F]^+$	-0.890
MR103	301.0780	301.0783	$[C_{16}H_{11}N_2O_2F_2]^+$	-0.899
MR106	267.0574	267.0576	$[C_{12}H_9N_2O_3F_2]^+$	-0.581

Figure 1: Mass Spectrum of MR105 with Isotopic Model

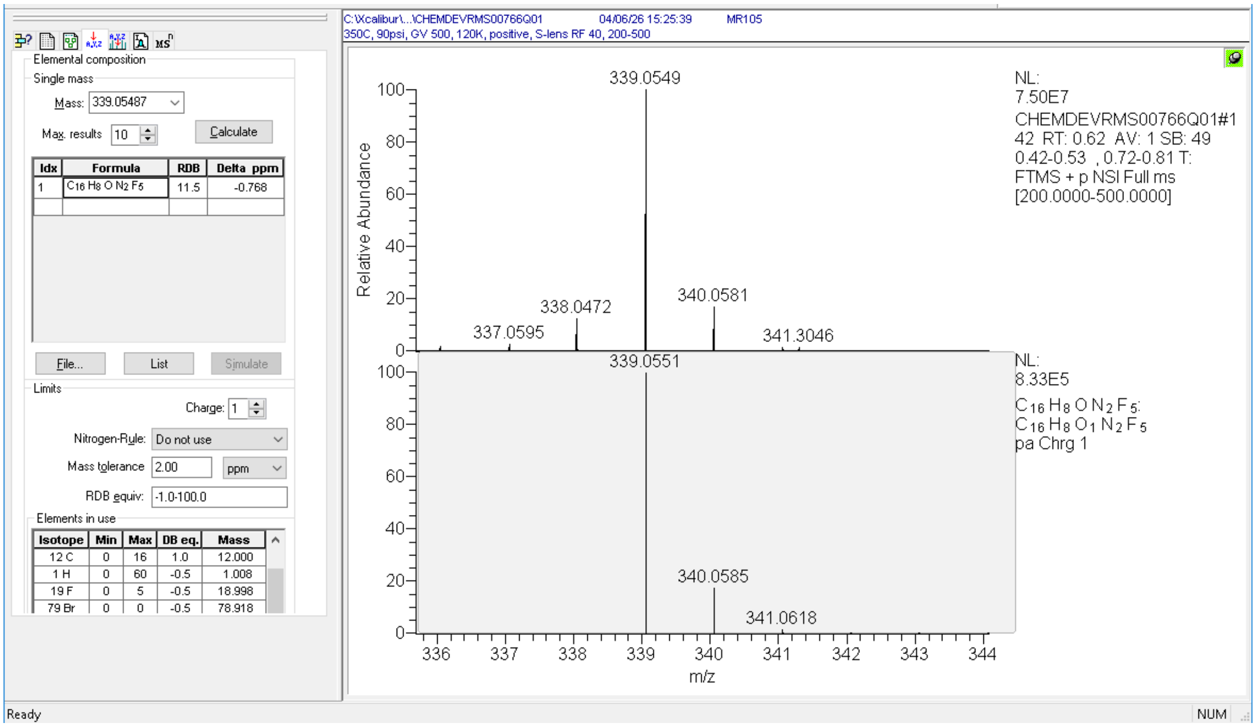


Figure 2: Mass Spectrum of MR56 with Isotopic Model

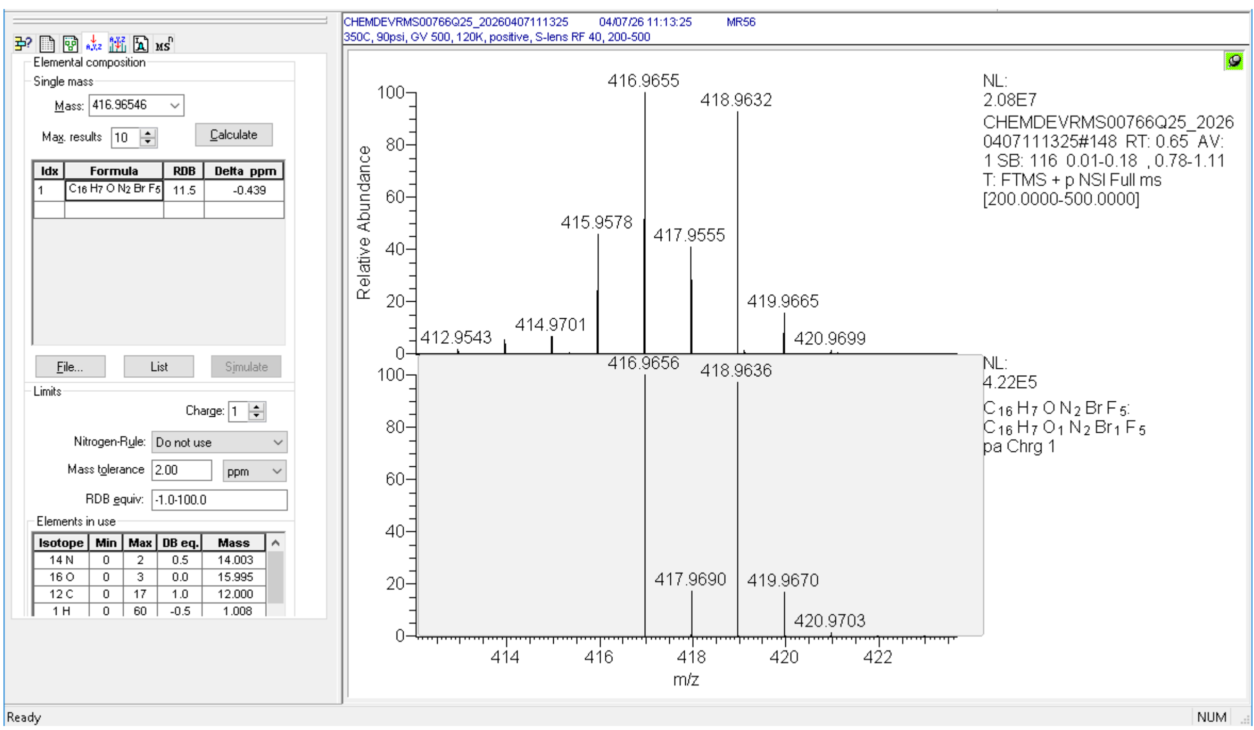


Figure 3: Mass Spectrum of MR62 with Isotopic Model

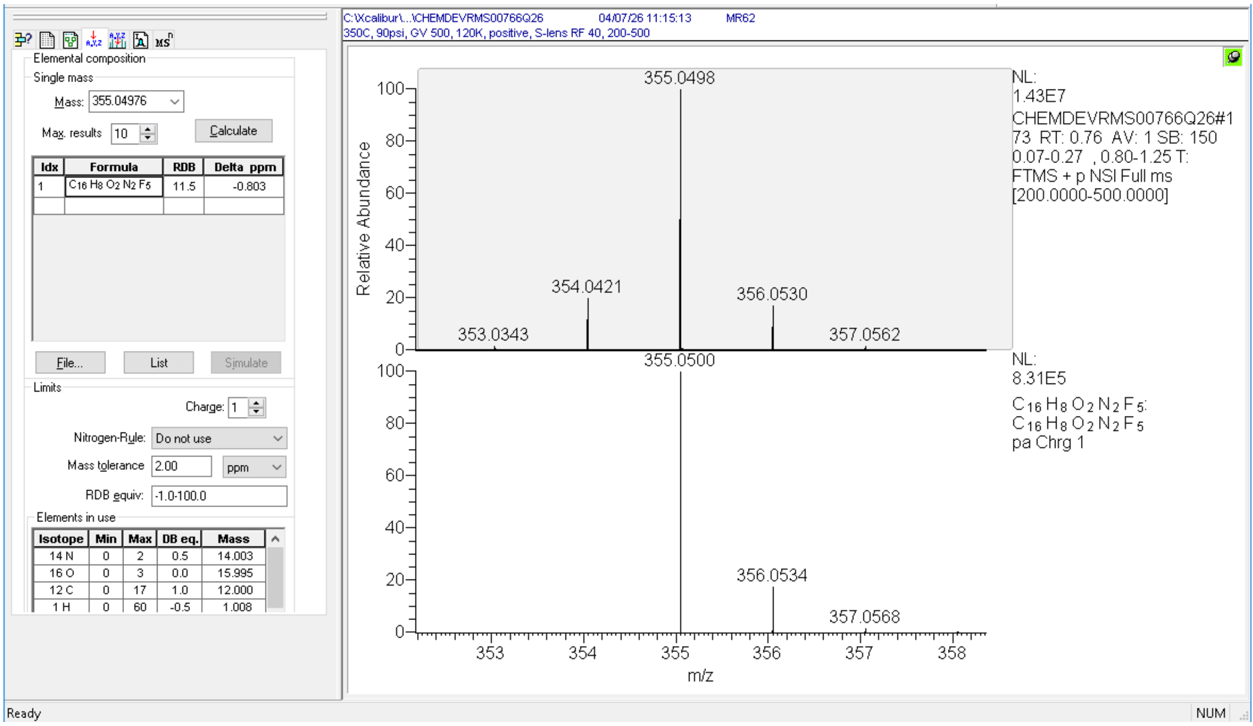


Figure 4: Mass Spectrum of MR63 with Isotopic Model

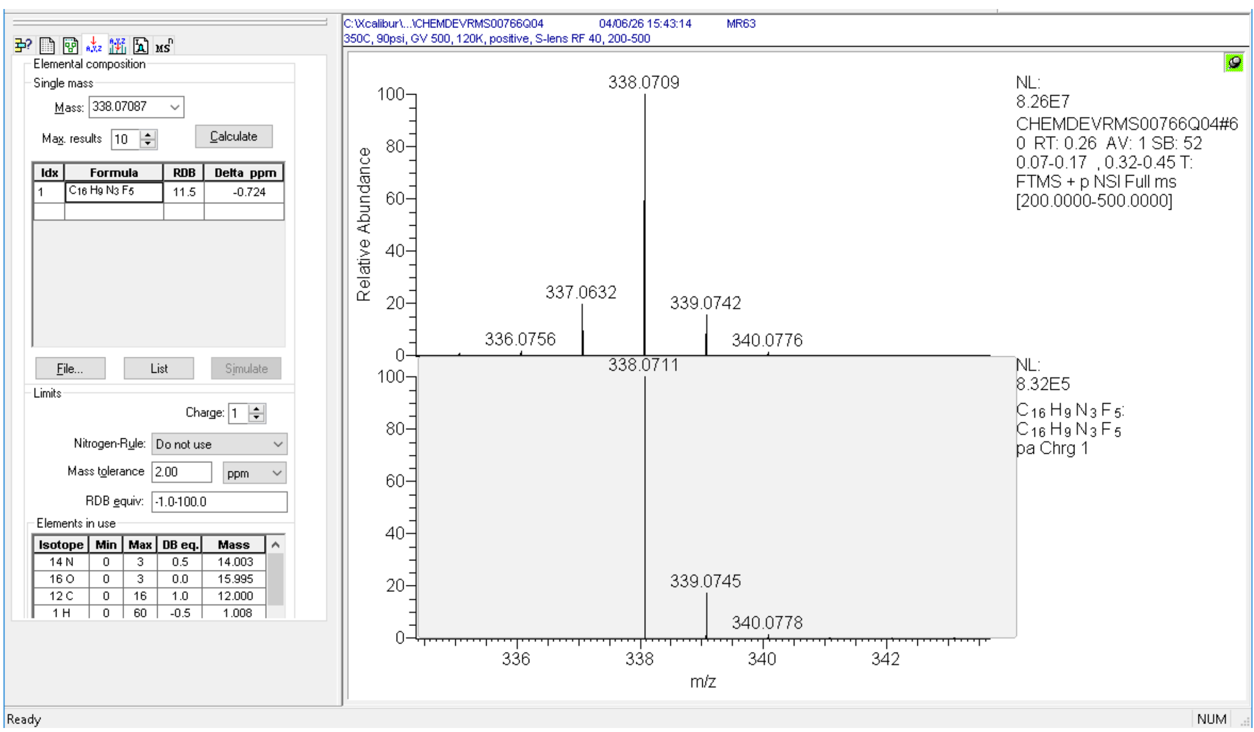


Figure 5: Mass Spectrum of MR89 with Isotopic Model

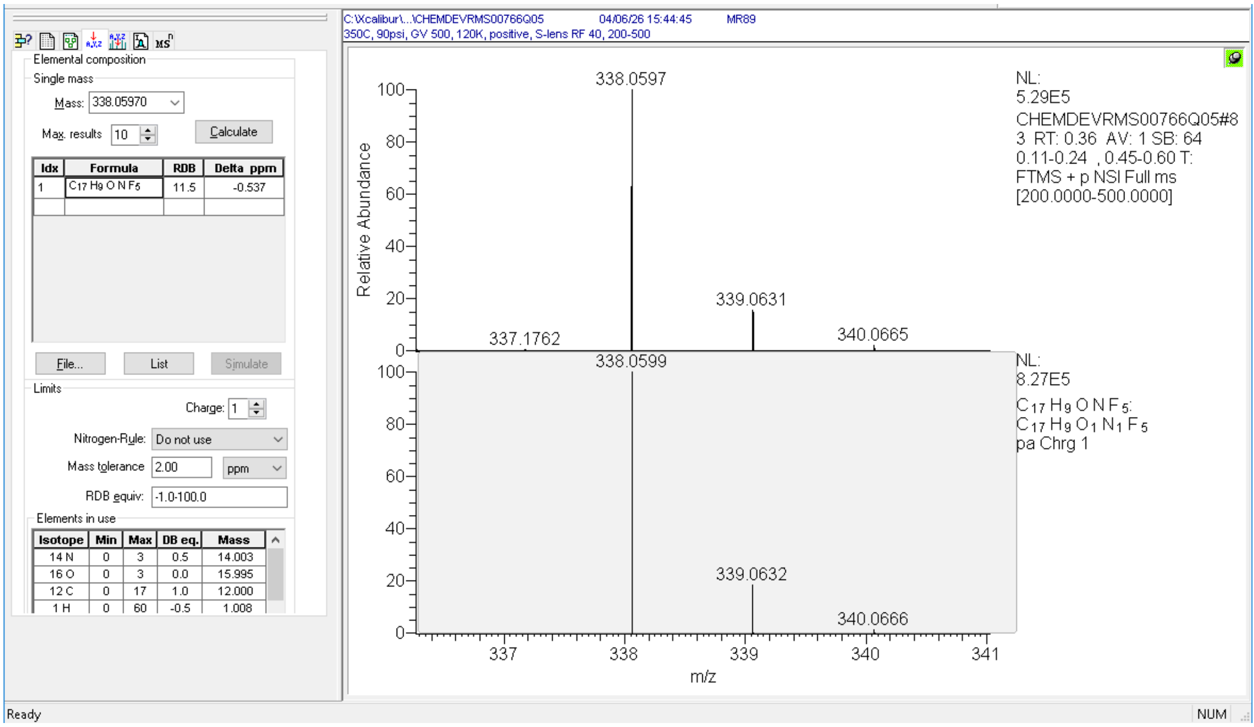


Figure 6: Mass Spectrum of MR65 with Isotopic Model

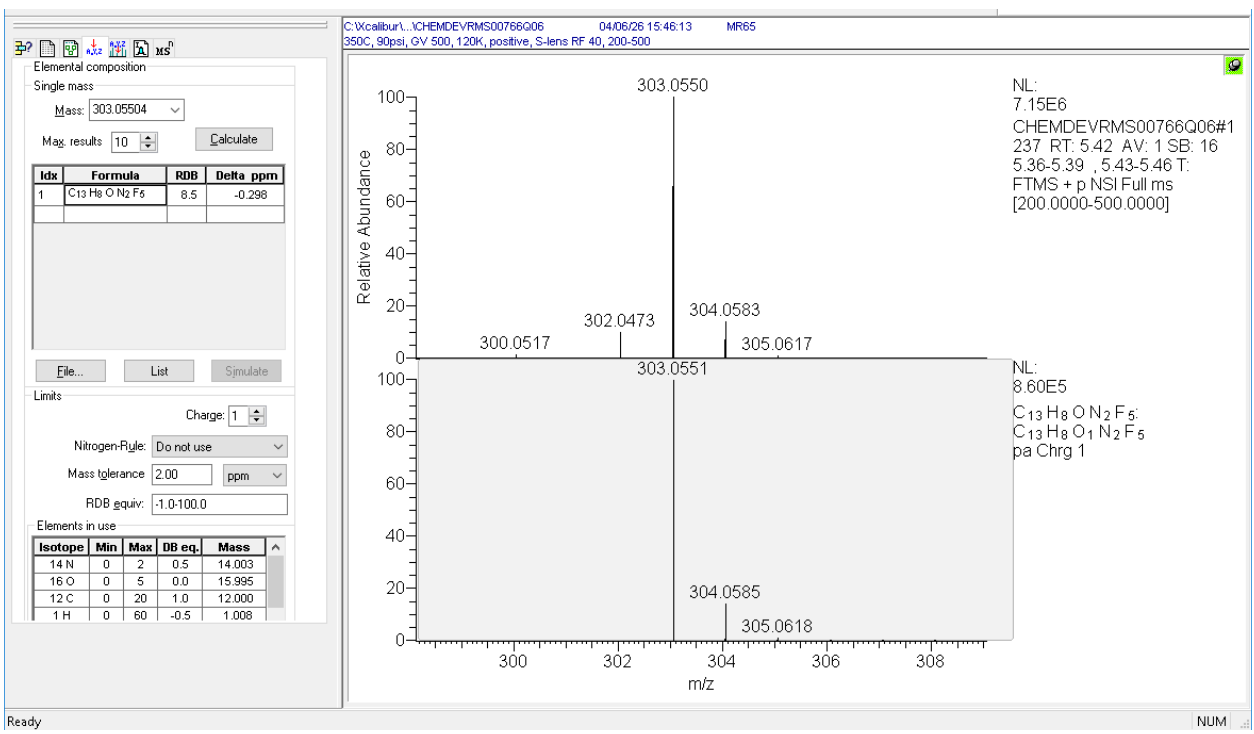


Figure 7: Mass Spectrum of MR64 with Isotopic Model

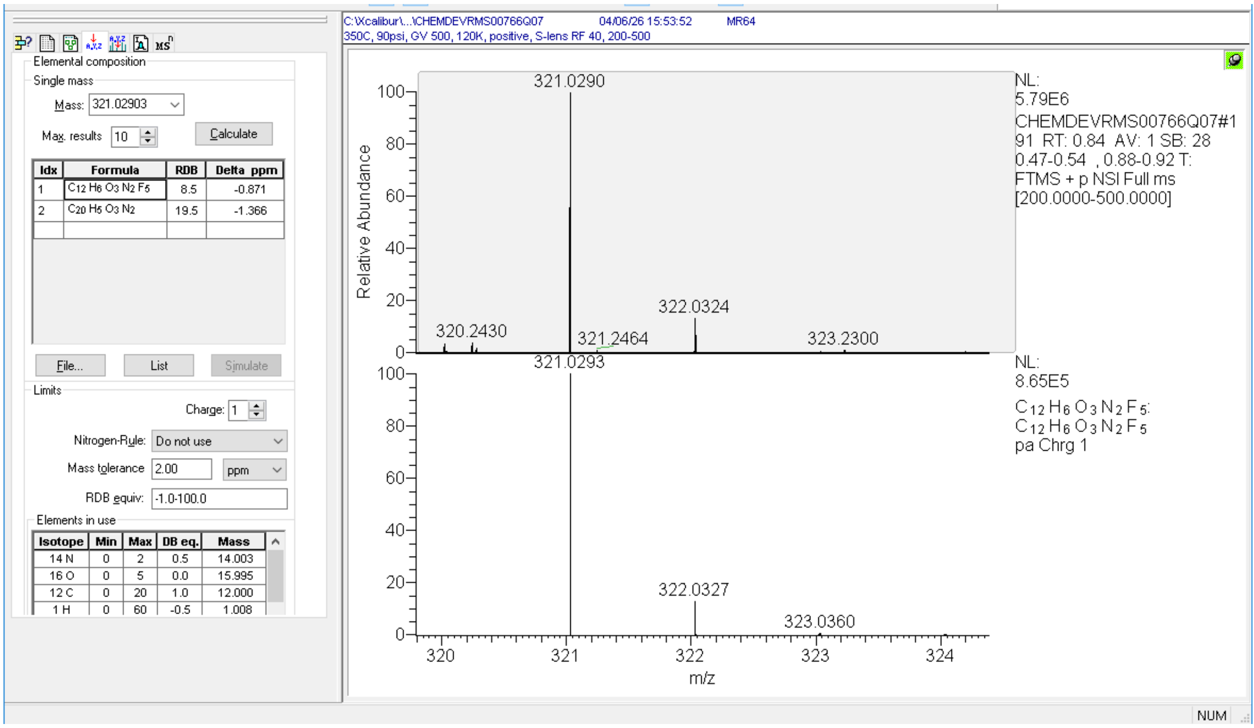


Figure 8: Mass Spectrum of MR73 with Isotopic Model

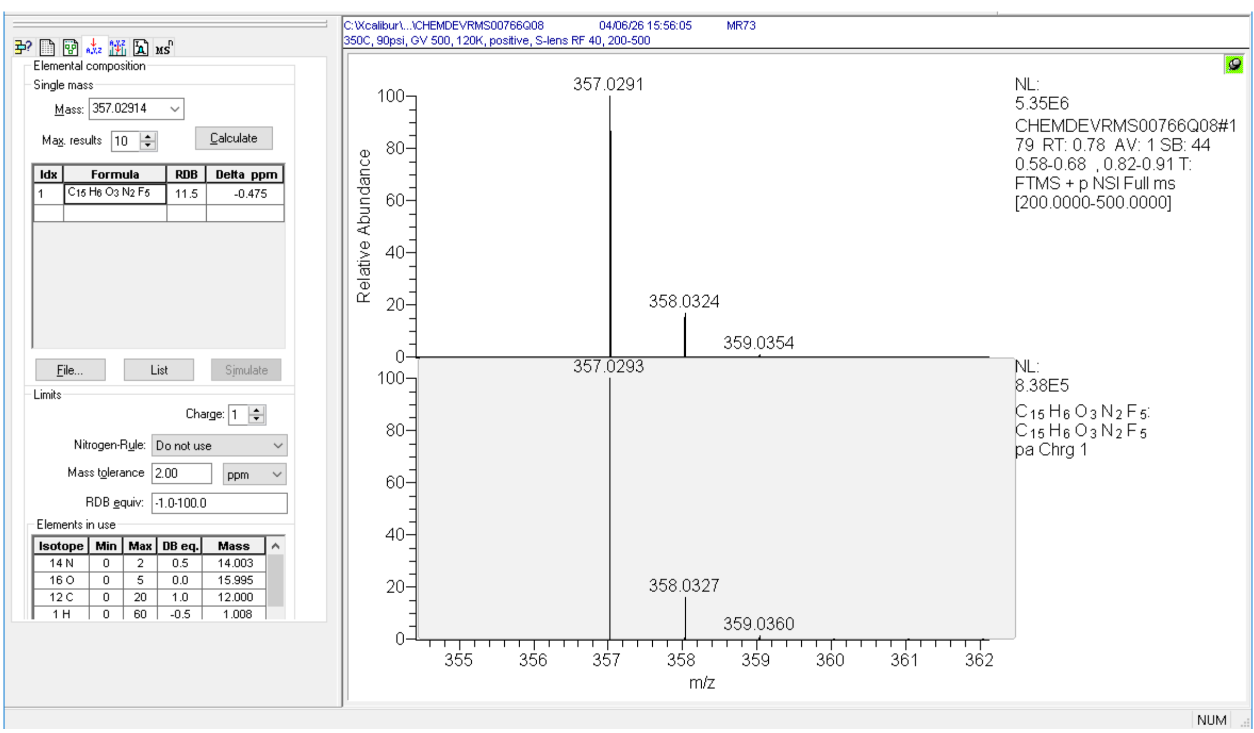


Figure 9: Mass Spectrum of MR99 with Isotopic Model

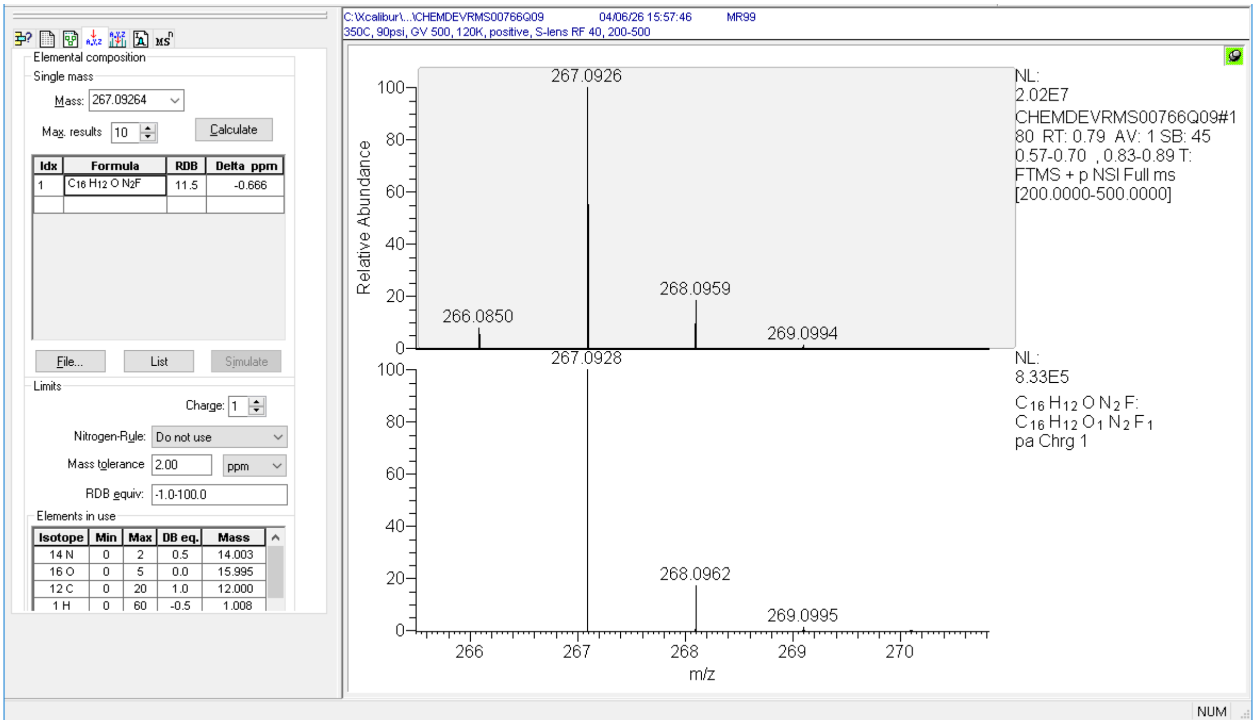


Figure 10: Mass Spectrum of MR92 with Isotopic Model

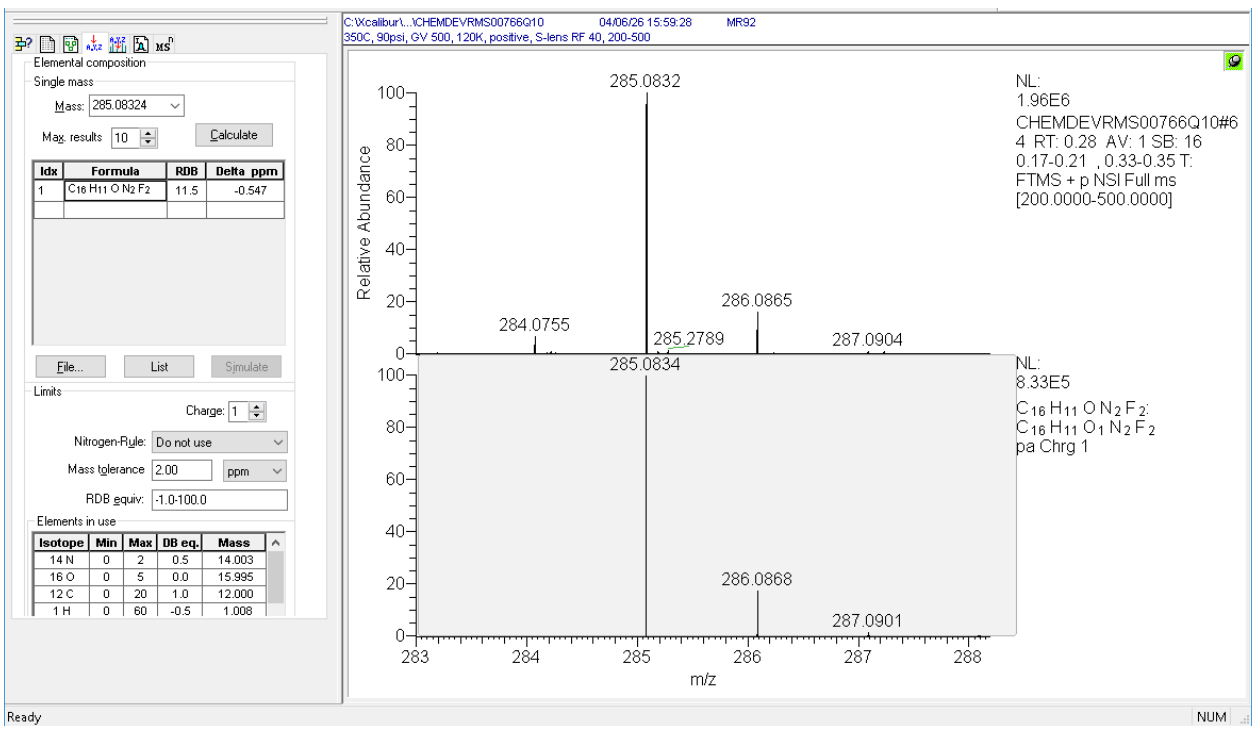


Figure 11: Mass Spectrum of MR88 with Isotopic Model

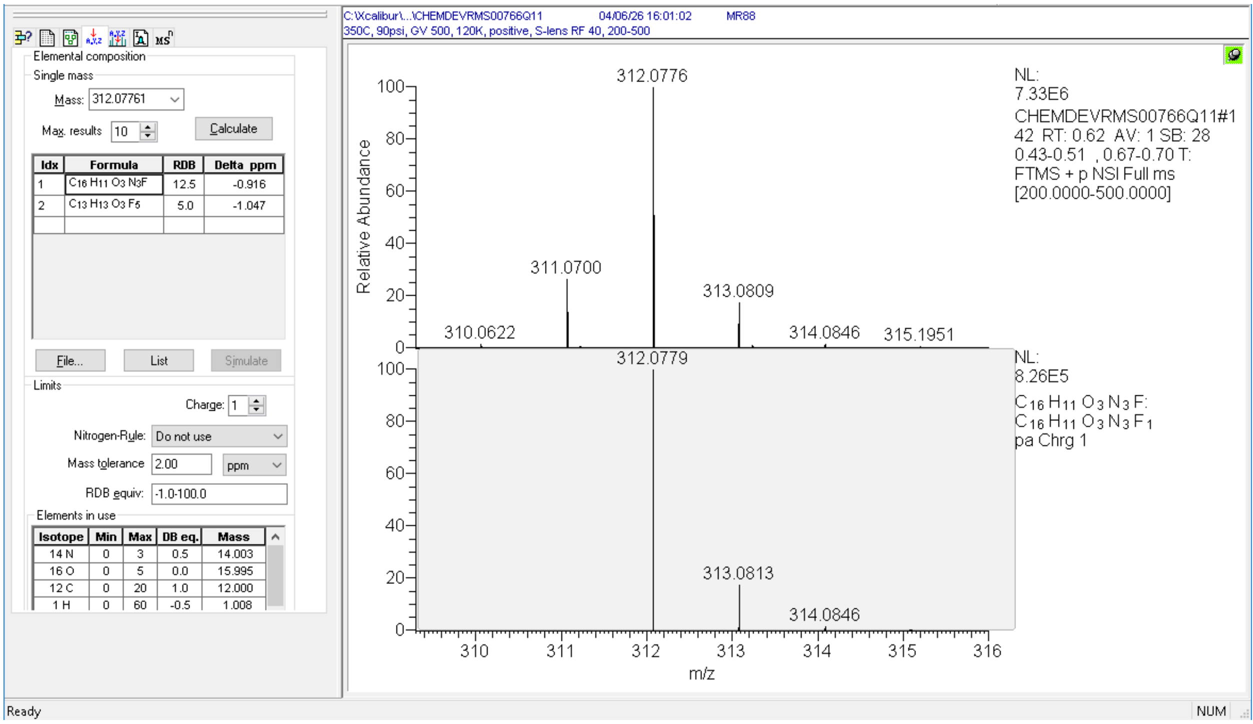


Figure 12: Mass Spectrum of MR116 with Isotopic Model

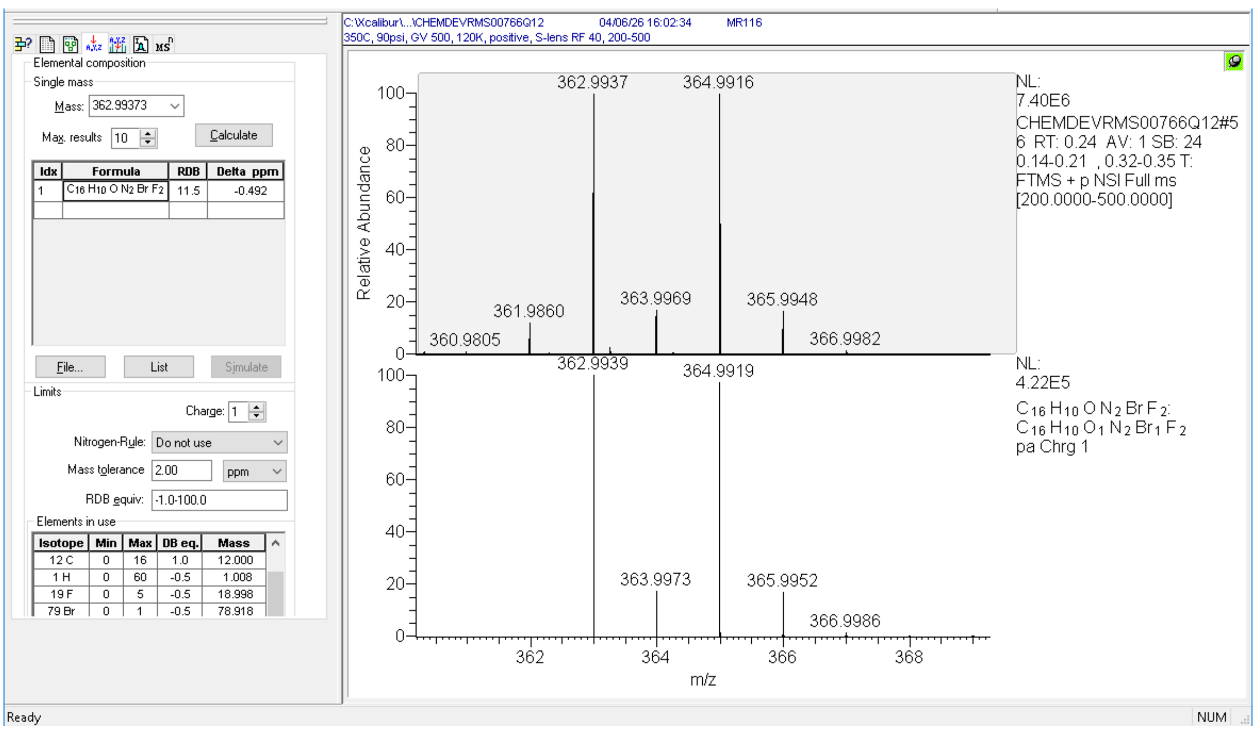


Figure 13: Mass Spectrum of MR111 with Isotopic Model

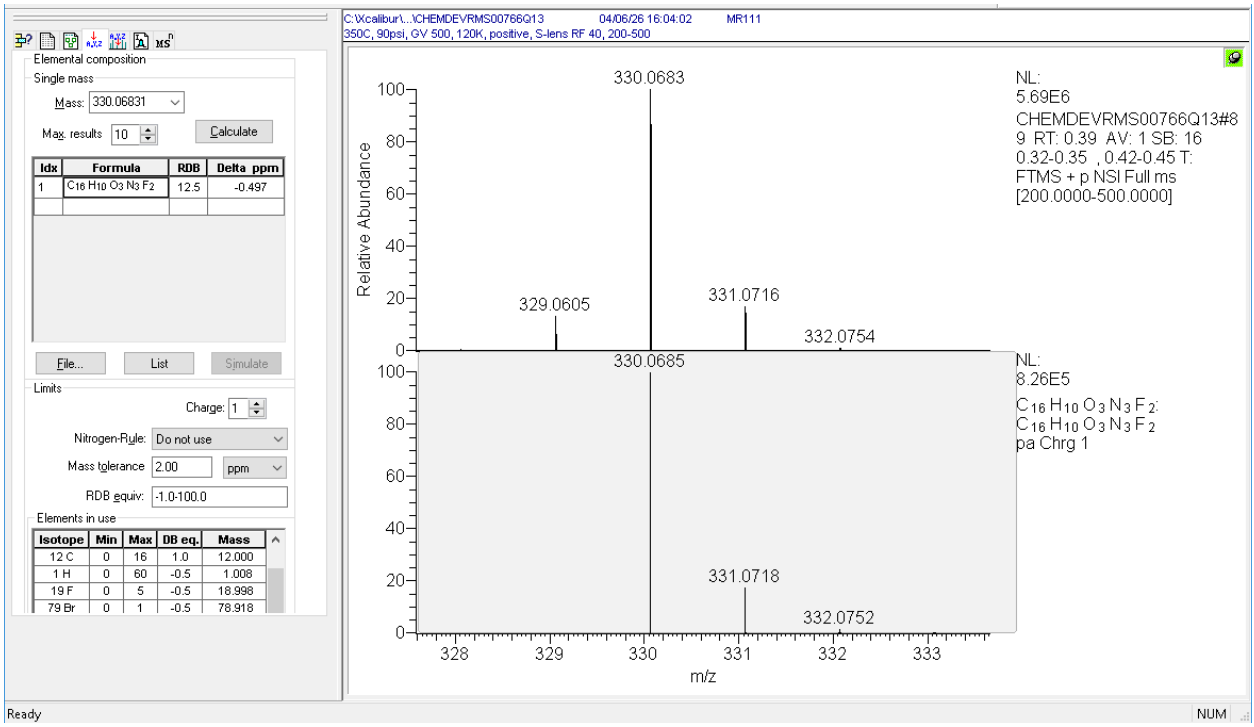


Figure 14: Mass Spectrum of MR113 with Isotopic Model

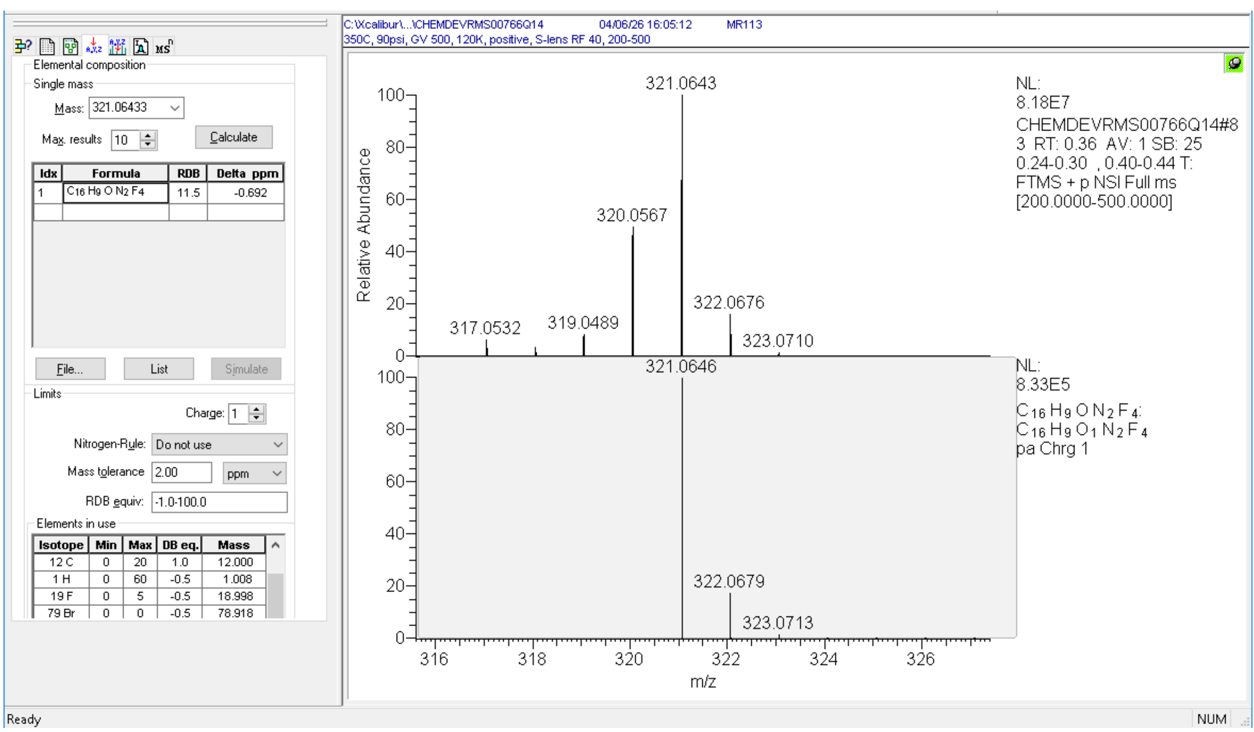


Figure 15: Mass Spectrum of MR94 with Isotopic Model

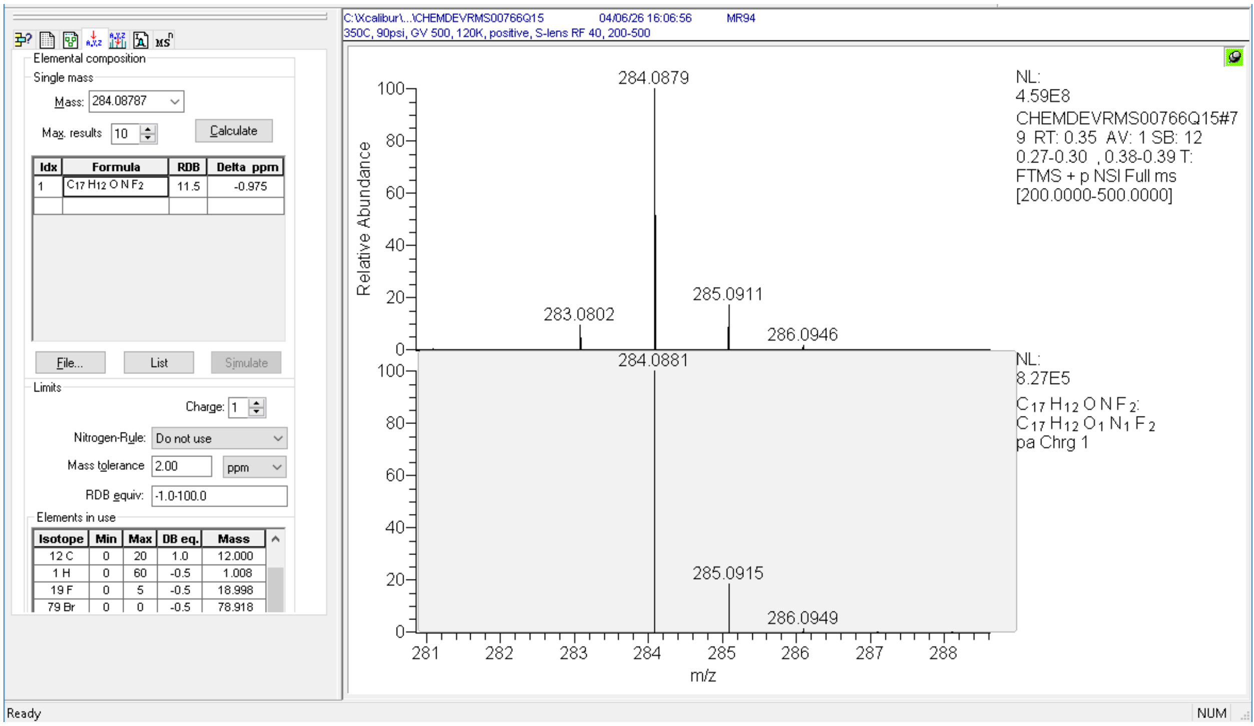


Figure 16: Mass Spectrum of MR108 with Isotopic Model

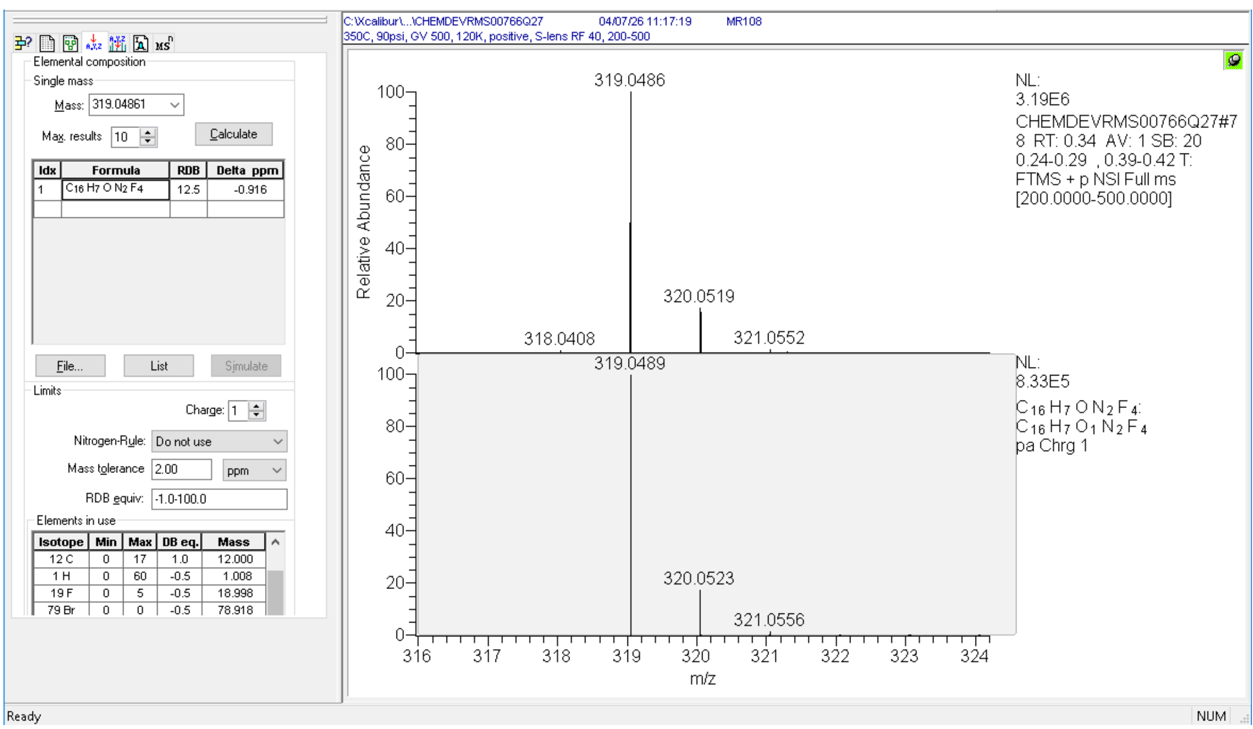


Figure 17: Mass Spectrum of MR101 with Isotopic Model

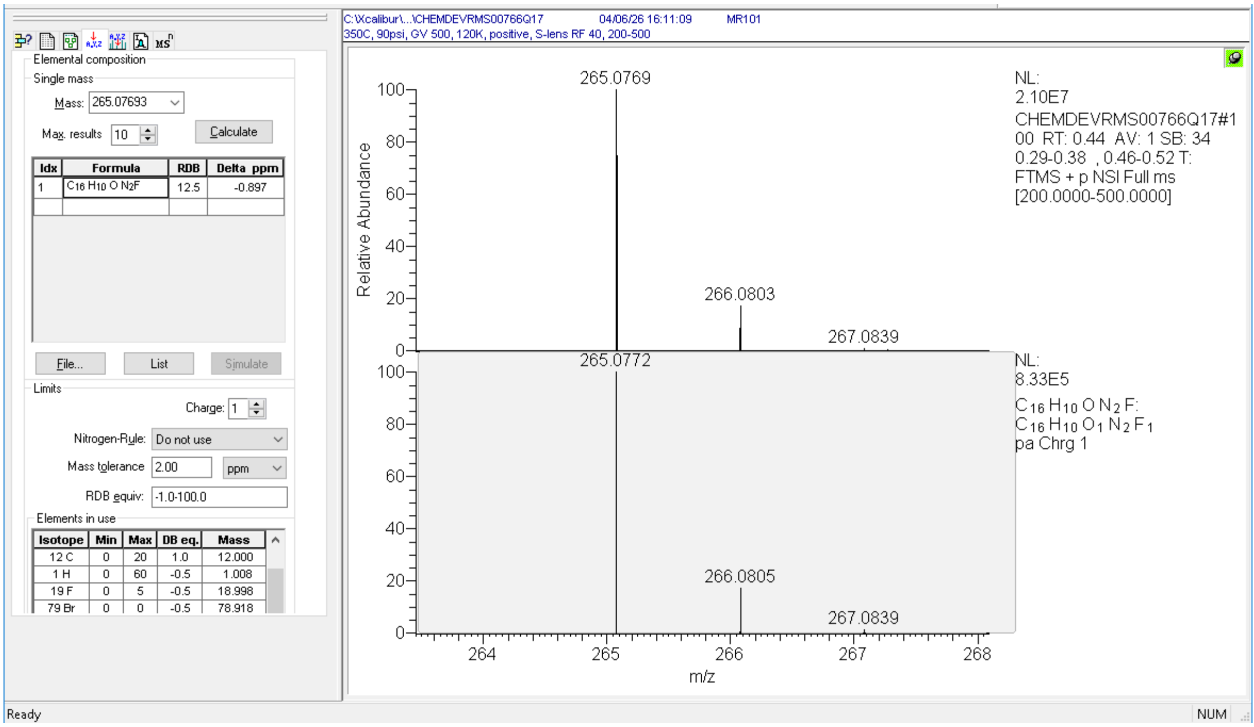


Figure 18: Mass Spectrum of MR59 with Isotopic Model

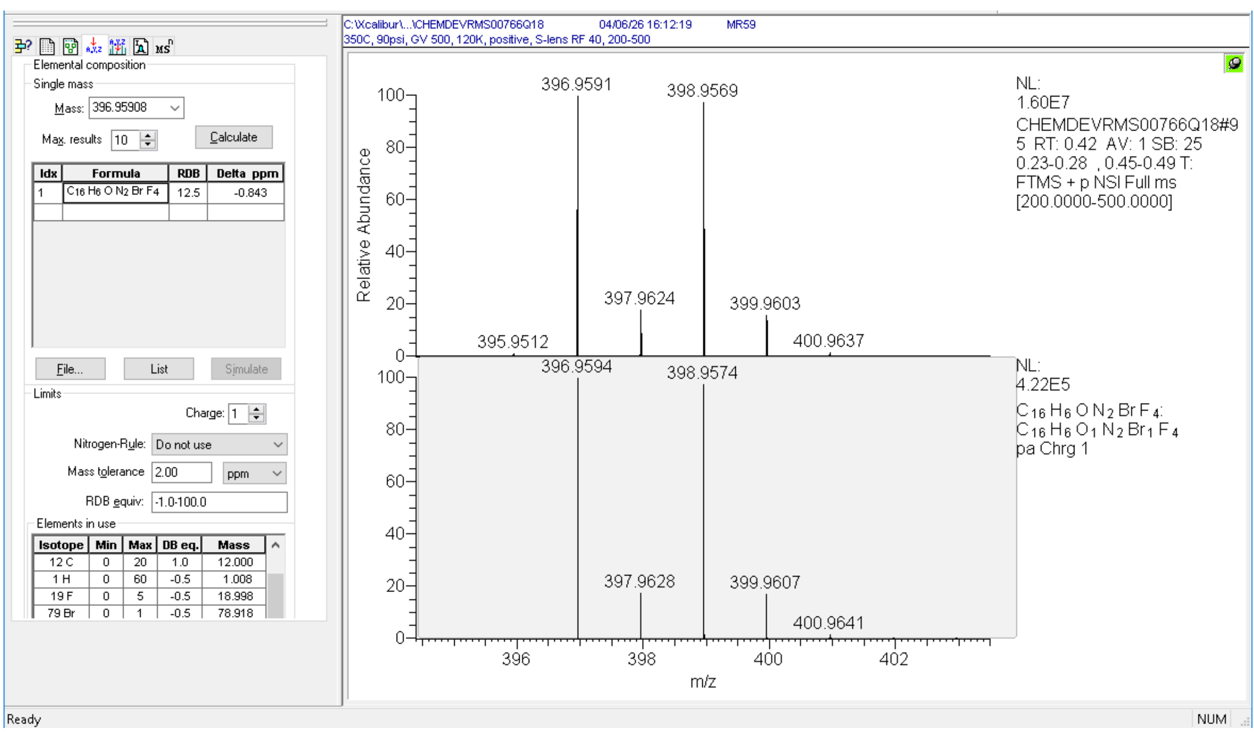


Figure 19: Mass Spectrum of MR96 with Isotopic Model

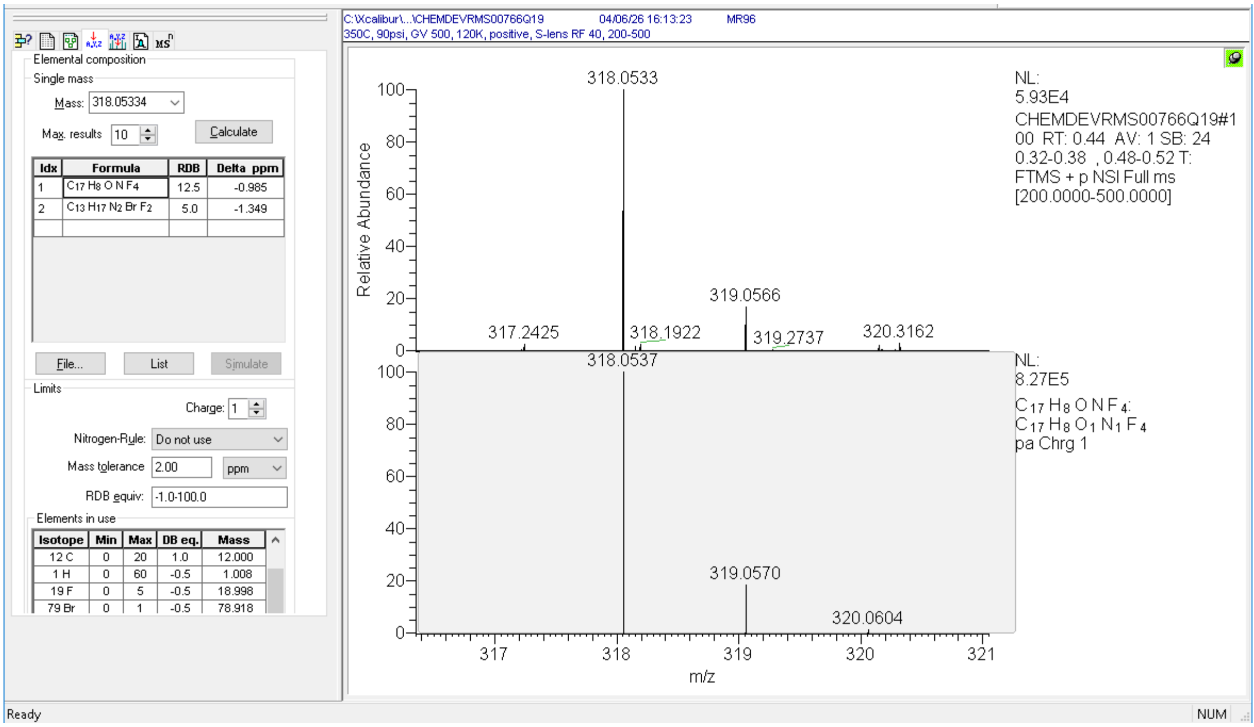


Figure 20: Mass Spectrum of MR115 with Isotopic Model

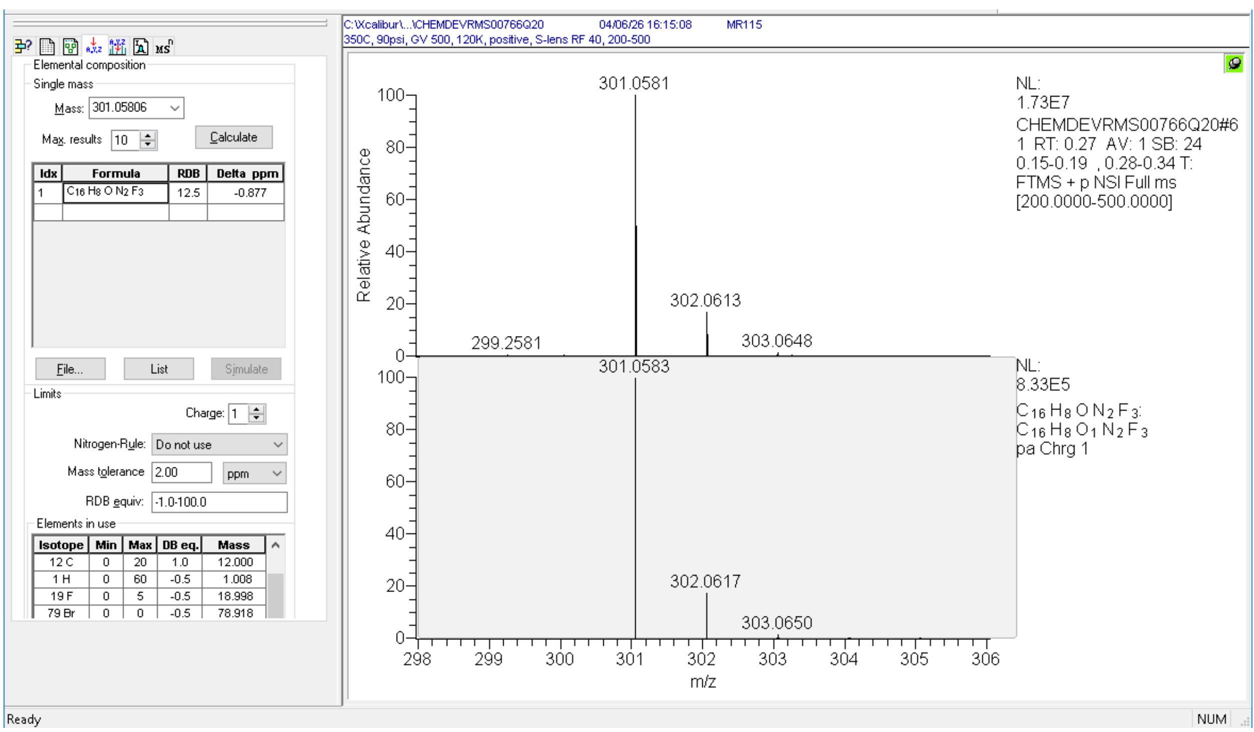


Figure 21: Mass Spectrum of MR120 with Isotopic Model

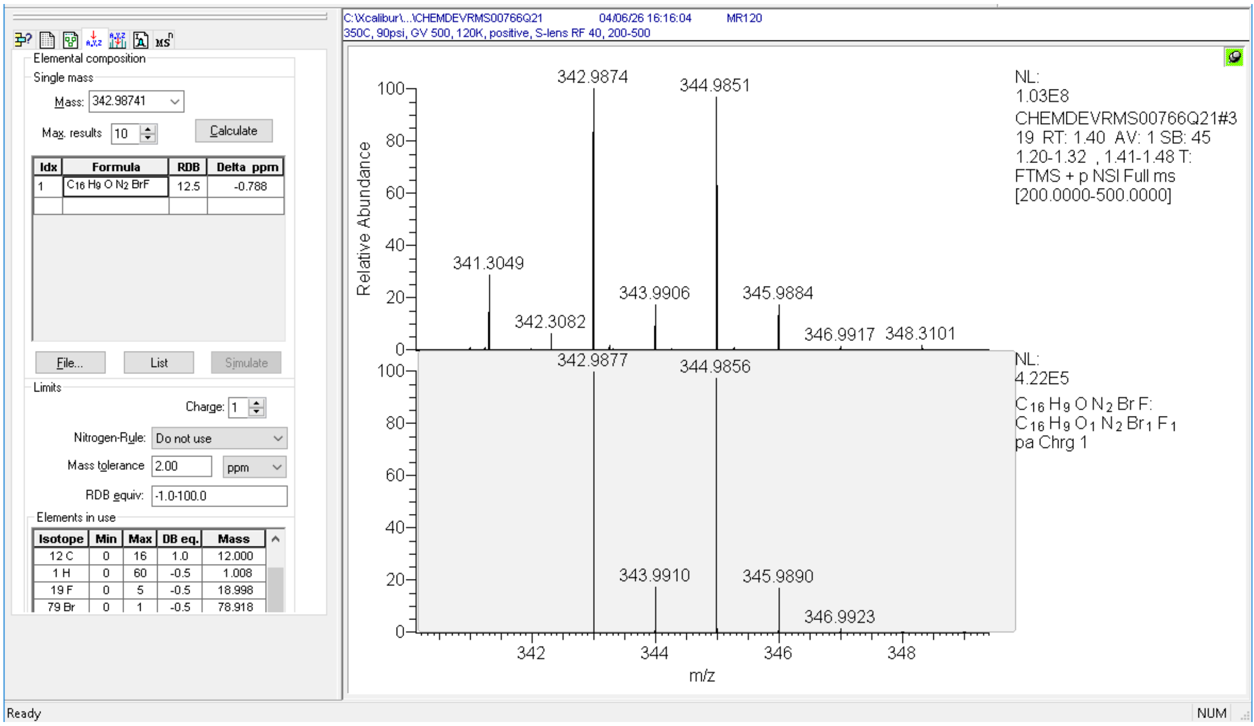


Figure 22: Mass Spectrum of MR119 with Isotopic Model

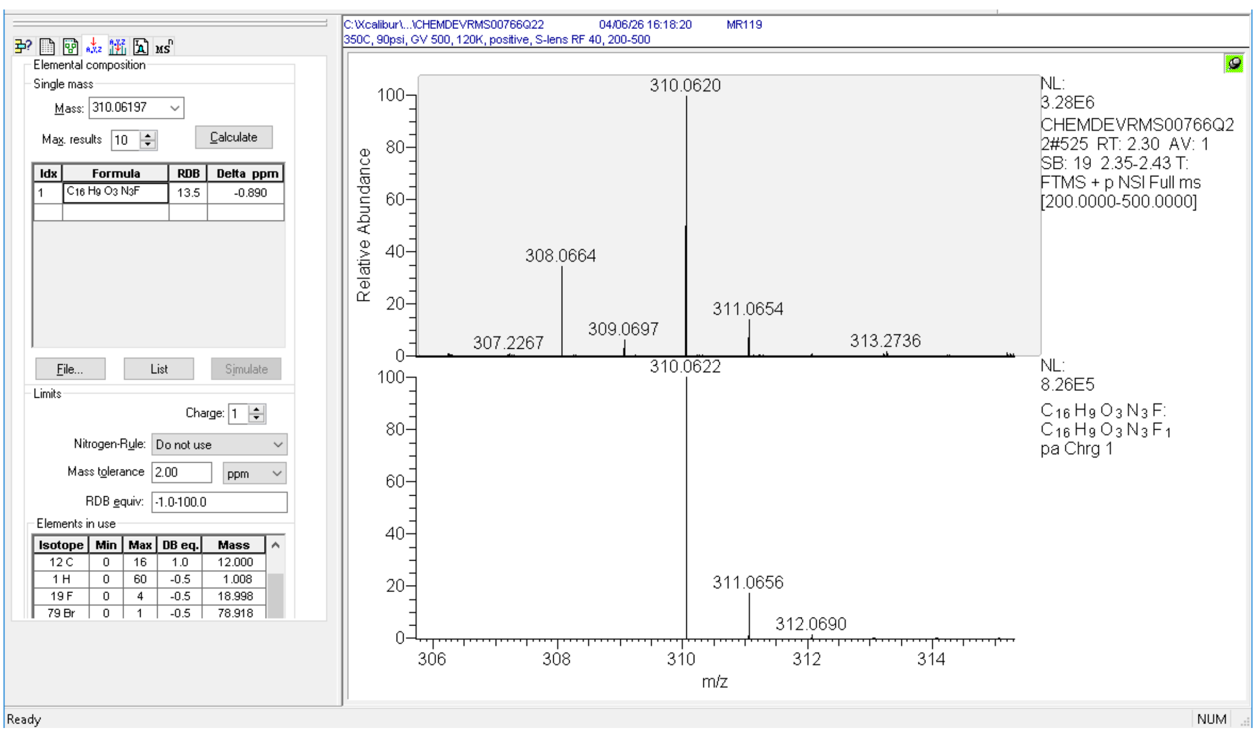


Figure 23: Mass Spectrum of MR103 with Isotopic Model

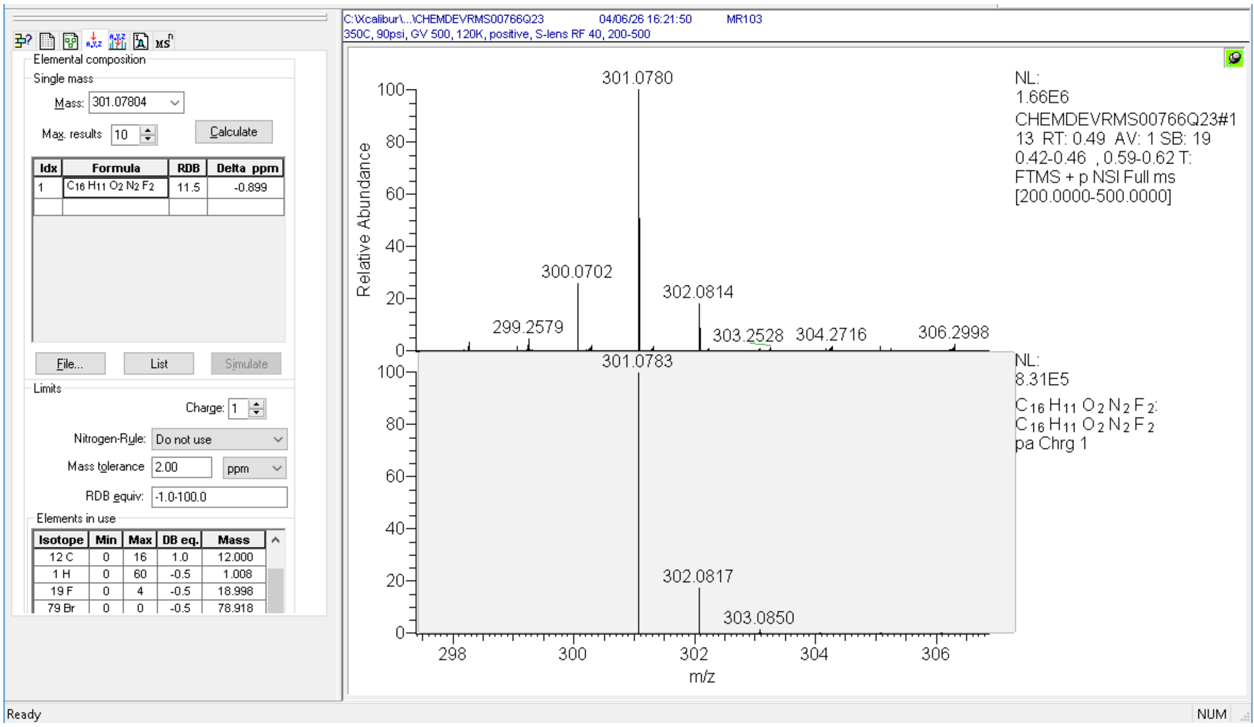
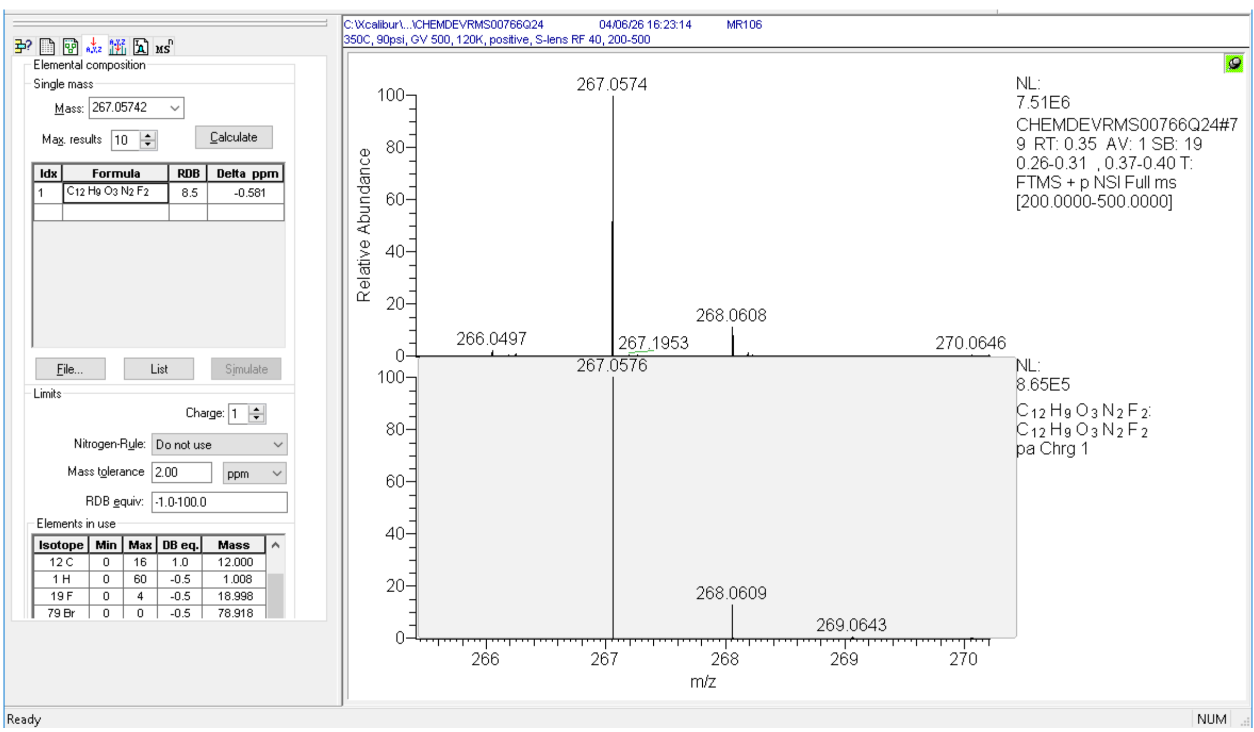


Figure 24: Mass Spectrum of MR106 with Isotopic Model



Appendix 1: MS Instruments and Methods

Instruments	
Thermo Q Exactive HF Mass Spectrometer	SN05260L
Ion Sense DART Controller SVP100	SVP00058
Ion Sense DART SVP Source	SVPS00070
DART Parameters	
Ionization Mode	Positive Ion DART
Resolution Power	120,000
Heater Temperature	350ºC
Capillary Temperature	275°C
He Tank Pressure	90 psi
Grid Voltage	500 V
S Lens RF Level	40 V
Scan Range	200 to 500 a.m.u.

Appendix 2: RAS Form

Boehringer Ingelheim

MS #: 26-018
RMS00766

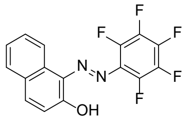
Material and Analytical Sciences – Request for Mass Spectrometry Support

Requester: Jon Reeves	Sample Description: 24 samples of oxadiazepines	Lot No(s). See attached sheet.
Work Requested: HRMS		Storage Conditions (circle) RT | R | F | CF | D | PL
		Date Sampled/Sampler Initials
		Received by/Date (testing lab)

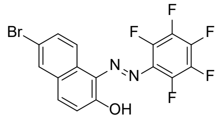
Additional Sample Information	Individual Sample Identity	Sample ID #
Quantity of Sample: (g, mg, mL) 2-20 mg	See attached ChemDraw file	
Project Code: BD70378 (BRAF Protac)		
Category: default BIEL 3B		
non-GMP		

Results:

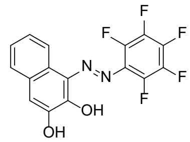
Appendix 3: Structures of 24 Oxadiazepine Compounds



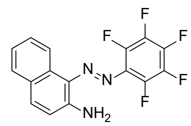
Lot: MR105
Formula: $C_{16}H_7F_5N_2O$
Exact Mass: 338.04785



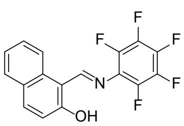
Lot: MR56
Formula: $C_{16}H_6BrF_5N_2O$
Exact Mass: 415.95837



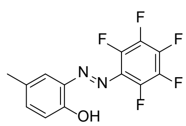
Lot: MR62
Formula: $C_{16}H_7F_5N_2O_2$
Exact Mass: 354.04277



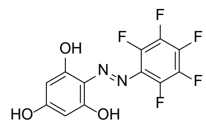
Lot: MR63
Formula: $C_{16}H_8F_5N_3$
Exact Mass: 337.06384



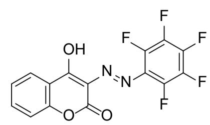
Lot: MR89
Formula: $C_{17}H_8F_5NO$
Exact Mass: 337.05260



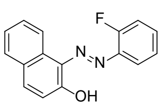
Lot: MR65
Formula: $C_{13}H_7F_5N_2O$
Exact Mass: 302.04785



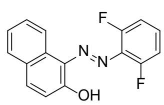
Lot: MR64
Formula: $C_{12}H_5F_5N_2O_3$
Exact Mass: 320.02203



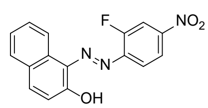
Lot: MR73
Formula: $C_{15}H_5F_5N_2O_3$
Exact Mass: 356.02203



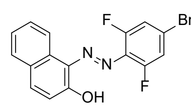
Lot: MR99
Formula: $C_{16}H_{11}FN_2O$
Exact Mass: 266.08554



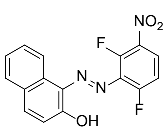
Lot: MR92
Formula: $C_{16}H_{10}F_2N_2O$
Exact Mass: 284.07612



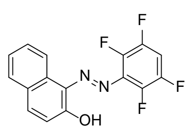
Lot: MR88
Formula: $C_{16}H_{10}FN_3O_3$
Exact Mass: 311.07062



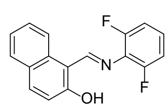
Lot: MR116
Formula: $C_{16}H_9BrF_2N_2O$
Exact Mass: 361.98663



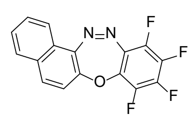
Lot: MR111
Formula: $C_{16}H_9F_2N_3O_3$
Exact Mass: 329.06120



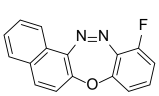
Lot: MR113
Formula: $C_{16}H_8F_4N_2O$
Exact Mass: 320.05728



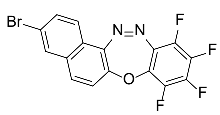
Lot: MR94
Formula: $C_{17}H_{11}F_2NO$
Exact Mass: 283.08087



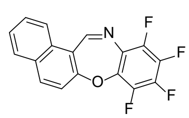
Lot: MR108
Formula: $C_{16}H_6F_4N_2O$
Exact Mass: 318.04163



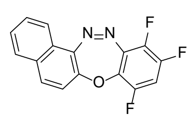
Lot: MR101
Formula: $C_{16}H_9FN_2O$
Exact Mass: 264.06989



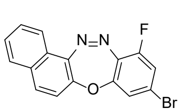
Lot: MR59
Formula: $C_{16}H_5BrF_4N_2O$
Exact Mass: 395.95214



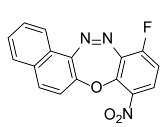
Lot: MR96
Formula: $C_{17}H_7F_4NO$
Exact Mass: 317.04638



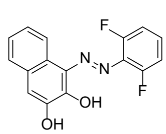
Lot: MR115
Formula: $C_{16}H_7F_3N_2O$
Exact Mass: 300.05105



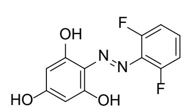
Lot: MR120
Formula: $C_{16}H_8BrFN_2O$
Exact Mass: 341.98040



Lot: MR119
Formula: $C_{16}H_8FN_3O_3$
Exact Mass: 309.05497



Lot: MR103
Formula: $C_{16}H_{10}F_2N_2O_2$
Exact Mass: 300.07103



Lot: MR 106
Formula: $C_{12}H_8F_2N_2O_3$
Exact Mass: 266.05030